

Homework 2 - ET4369 Nyquist Rate Data Converters

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1 Reconstruction (ZOH)

Given: Discrete-time signal $x(n)$ with $f_s = 125$ MHz, converted to continuous time using zero-order-hold (ZOH) pulses.

1.1 Part (a): Amplitude Envelopes

For a rectangular hold pulse of width T_p , the ZOH output spectrum is the sampled spectrum multiplied by the amplitude envelope:

$$|H_p(f)| = \frac{T_p}{T_s} \cdot |\text{sinc}(f \cdot T_p)| = \frac{T_p}{T_s} \cdot \left| \frac{\sin(\pi f T_p)}{\pi f T_p} \right| \quad (1)$$

where $T_s = 1/f_s = 8$ ns.

The three hold pulse widths and their DC gains:

T_p	T_p/T_s	Description
2 ns	0.25	Short RZ pulse ($\frac{1}{4}$ of period)
4 ns	0.50	Half-period RZ pulse
8 ns	1.00	Full-period NRZ ($T_p = T_s$)

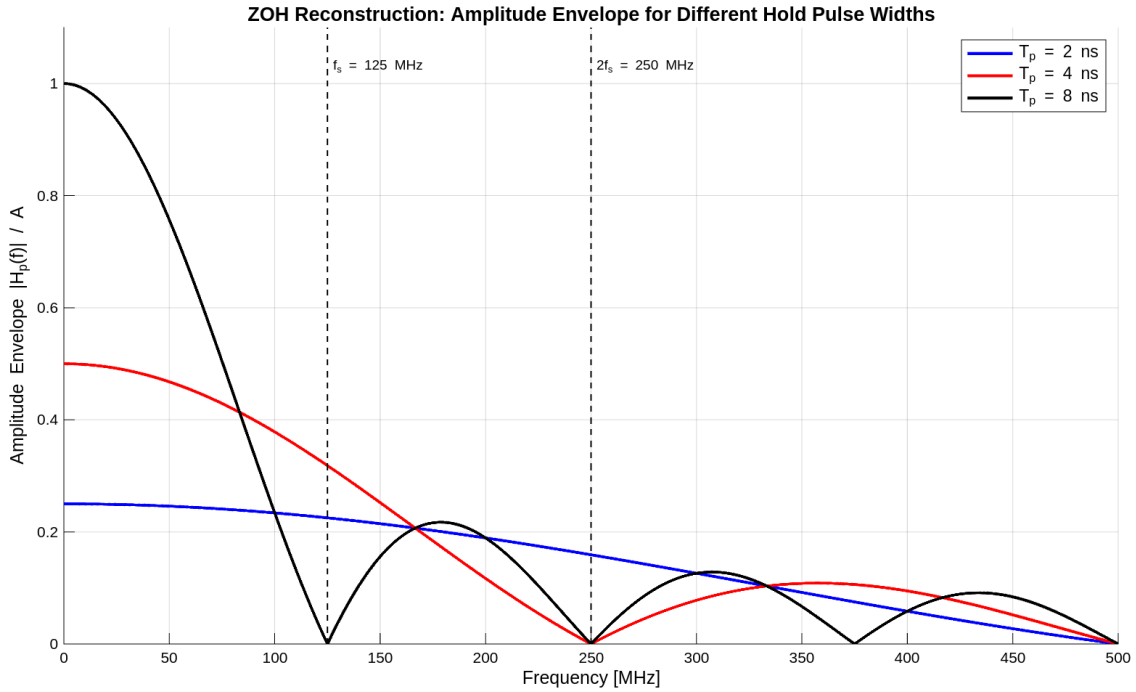


Figure 1: ZOH amplitude envelopes for three hold pulse widths.

Observations:

- The 8 ns (NRZ) case has the highest DC gain (1.0) but the steepest sinc roll-off, with a null at $f_s = 125$ MHz. This provides the strongest suppression of spectral images near f_s , but also the most in-band signal droop.
- The 2 ns case has the flattest response (first null at $1/T_p = 500$ MHz), but only 0.25 DC gain. Images are poorly suppressed.
- The 4 ns case is intermediate: first null at 250 MHz, DC gain of 0.5.

1.2 Part (b): Output Tones for 8 ns Hold, 30 MHz Sine Input

With $T_p = 8$ ns (NRZ, $T_p = T_s$), $f_{\text{in}} = 30$ MHz, and $f_s = 125$ MHz, the sampled signal contains spectral images at $f = f_{\text{in}} + k \cdot f_s$ for all integers k . Folding to positive frequencies, the tones up to 250 MHz are:

Tone	Frequency	Origin	$f \cdot T_p$	$\text{sinc}(f \cdot T_p)$	Amplitude/ A
1	30 MHz	f_{in}	0.24	0.9079	0.9079
2	95 MHz	$f_s - f_{\text{in}}$	0.76	0.2867	0.2867
3	155 MHz	$f_s + f_{\text{in}}$	1.24	0.1757	0.1757
4	220 MHz	$2f_s - f_{\text{in}}$	1.76	0.1238	0.1238

Table 1: Output tones for 8 ns NRZ hold with 30 MHz sine input.

Each tone's amplitude (normalized to A) equals the sinc envelope evaluated at that frequency:

$$\text{Amplitude}(f) = |\text{sinc}(f \cdot T_p)| = \frac{|\sin(\pi f T_p)|}{\pi f T_p} \quad (2)$$

since $T_p/T_s = 1$ for the NRZ case.

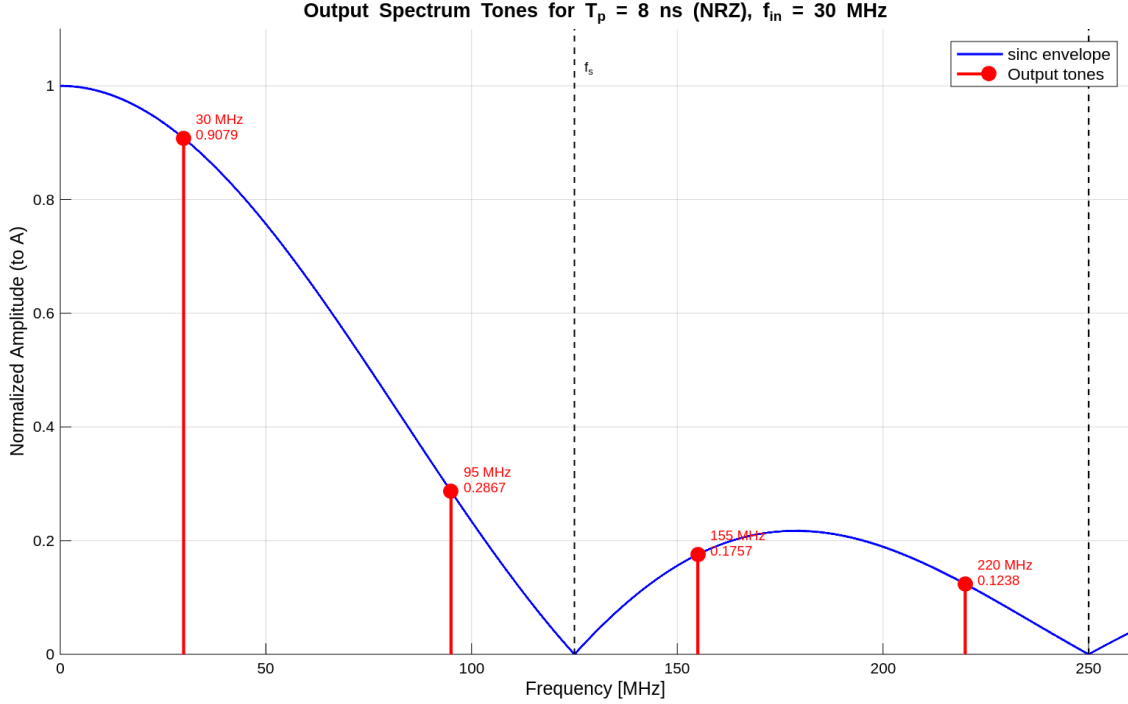


Figure 2: Output spectrum tones for $T_p = 8$ ns (NRZ), $f_{\text{in}} = 30$ MHz.

Observations:

- The fundamental at 30 MHz suffers about 9.2% attenuation (sinc droop) — from A to $0.908A$.
- The first image at 95 MHz ($= f_s - f_{\text{in}}$) is the strongest unwanted tone at $0.287A$ (-10.0 dB relative to the fundamental).
- Images at 155 MHz and 220 MHz are further suppressed to $0.176A$ and $0.124A$ respectively.
- A reconstruction filter is needed to suppress these image tones and optionally compensate the sinc droop.

2 Segmented DAC Design — Paper Review

Paper: C.-H. Lin and B. Bult, “A 10-b, 500-MSample/s CMOS DAC in 0.6 mm²,” *IEEE J. Solid-State Circuits*, vol. 33, no. 12, pp. 1948–1958, Dec. 1998.

2.1 What application is this DAC intended for?

The DAC is designed for **cable modem headend transmitters** — a frequency-domain application where the DAC must synthesize high-quality sinusoidal tones across the cable modem upstream band (5–65 MHz).

In a cable modem headend, the transmitter generates multiple QAM-modulated carriers simultaneously. The DAC converts the digitally synthesized waveform to analog at high speed (f_s up to 500 MS/s). Since the application is in the frequency domain, **spectral purity (SFDR)** is the primary performance metric rather than time-domain linearity alone.

Table 2: Key specifications of the Lin & Bult DAC.

Parameter	Value
Resolution	10 bits
Sampling rate	up to 500 MS/s
SFDR	> 60 dB ($f_s \leq 200$ MS/s)
DNL / INL	0.1 LSB / 0.2 LSB
Power	125 mW at 500 MS/s
Area	0.6 mm ²
Technology	0.35 μ m, 1P4M, 3.3 V standard digital CMOS

2.2 Architecture: 8+2 Segmented Current-Steering DAC

The DAC uses a segmented architecture with **8 thermometer-coded MSBs** and **2 binary-weighted LSBs**:

- **Thermometer coding** (MSBs) ensures monotonicity, reduces glitch energy, and improves DNL — critical for spectral purity.
- **Binary weighting** (LSBs) reduces decoder complexity and area for the least significant bits where mismatch impact is small.
- The segmentation point is optimized so that the analog area (current source array) roughly equals the digital area (thermometer decoder), minimizing total die area (see Fig. 9 of the paper).

For cable modem headend use, harmonic distortion and intermodulation products from the DAC appear as spurious tones in adjacent channels, degrading system performance. The > 60 dB SFDR requirement ensures these spurs are sufficiently suppressed for reliable QAM transmission.

3 DAC Area Optimization

Task: Re-optimize the segmented DAC for a 65-nm technology with different process parameters.

3.1 Given Parameters

Parameter	Description	Value
B	Total resolution	12 bits
INL_{spec}	Worst case INL	$< 0.5 \text{ LSB}$
DNL_{spec}	Worst case DNL	$< 0.5 \text{ LSB}$
Y	Target yield	95%
A_{decode}	Decoder area	$2^{B_t} \cdot 2000 \mu\text{m}^2$
A_{total}	Total DAC area	$2^B \cdot A_{\text{unit}} + A_{\text{decode}}$
k_u	Matching parameter	$6\% \mu\text{m}$
σ_u	Unit element mismatch	$k_u / \sqrt{A_{\text{unit}}}$

3.2 Hand Calculations

Following the procedure from the Lin & Bult paper (Section IV–V, Table I, Fig. 9):

Worst-case DNL (at the major carry transition, where one thermometer element of 2^{B_b} unit elements switches ON and the binary section at full scale of $2^{B_b} - 1$ elements switches OFF):

$$\sigma_{\text{DNL},\text{max}} = \sqrt{2^{B_b+1} - 1} \cdot \sigma_u \quad \text{for } B_t \geq 1 \quad (3)$$

Worst-case INL (at midcode, independent of segmentation — see Fig. 8 of the paper):

$$\sigma_{\text{INL},\text{max}} = 0.5 \cdot \sqrt{2^B} \cdot \sigma_u = 32 \sigma_u \quad (4)$$

Following the paper’s procedure, we set $\sigma = \text{spec}$ (1σ confidence level):

DNL constraint:

$$\sqrt{2^{B_b+1} - 1} \cdot \sigma_u \leq 0.5 \quad \Rightarrow \quad A_{\text{unit,DNL}} = \frac{k_u^2 \cdot (2^{B_b+1} - 1)}{0.25} \quad (5)$$

INL constraint:

$$32 \sigma_u \leq 0.5 \quad \Rightarrow \quad \sigma_u \leq \frac{1}{64} \quad \Rightarrow \quad A_{\text{unit,INL}} = \left(\frac{0.06}{1/64} \right)^2 = 14.75 \mu\text{m}^2 \quad (6)$$

The INL constraint is independent of B_t and always requires $A_{\text{unit}} \geq 14.75 \mu\text{m}^2$.

3.3 Sweep Results

Table 3: Area sweep across integer B_t values.

B_t	B_b	$A_{\text{unit,DNL}}$ (μm^2)	A_{unit} (μm^2)	A_{analog} (μm^2)	A_{digital} (μm^2)	A_{total} (μm^2)	Active constraint
0	12	59.0	59.0	241,533	2,000	243,533	DNL
1	11	59.0	59.0	241,533	4,000	245,533	DNL
2	10	29.5	29.5	120,737	8,000	128,737	DNL
3	9	14.7	14.75	60,398	16,000	76,398	INL
4	8	7.4	14.75	60,398	32,000	92,398	INL
5	7	3.7	14.75	60,398	64,000	124,398	INL
6	6	1.8	14.75	60,398	128,000	188,398	INL
7	5	0.9	14.75	60,398	256,000	316,398	INL

3.4 Part (a)(i): Minimum Total Area

The minimum total area occurs at $\mathbf{B_t} = \mathbf{3}$, $B_b = 9$:

- $A_{\text{unit}} = 14.75 \mu\text{m}^2$ (set by INL; DNL nearly equal at $14.73 \mu\text{m}^2$)
- $A_{\text{analog}} = 2^{12} \times 14.75 = 60,398 \mu\text{m}^2$
- $A_{\text{digital}} = 2^3 \times 2000 = 16,000 \mu\text{m}^2$

$$A_{\text{total}} = 76,398 \mu\text{m}^2 \approx 0.076 \text{ mm}^2 \quad (7)$$

This is the crossover point where DNL and INL constraints meet. For $B_t < 3$, DNL dominates and requires a larger A_{unit} ; for $B_t > 3$, analog area is fixed but digital area grows exponentially.

3.5 Part (a)(ii): Optimal Point (Analog = Digital Area)

The “optimal point” from the paper is where $A_{\text{analog}} = A_{\text{digital}}$:

$$2^{B_t} \cdot 2000 = 60,398 \Rightarrow 2^{B_t} = 30.2 \Rightarrow B_t = 4.92 \quad (8)$$

The nearest integer is $B_t = 5$ ($B_b = 7$), where the analog/digital ratio is 0.94:

$$A_{\text{total}} = 60,398 + 64,000 = 124,398 \mu\text{m}^2 \approx 0.124 \text{ mm}^2 \quad (9)$$

This is about $1.6\times$ larger than the minimum-area design, but provides better THD performance due to more thermometer coding.

3.6 Part (b): Area vs Segmentation Plot

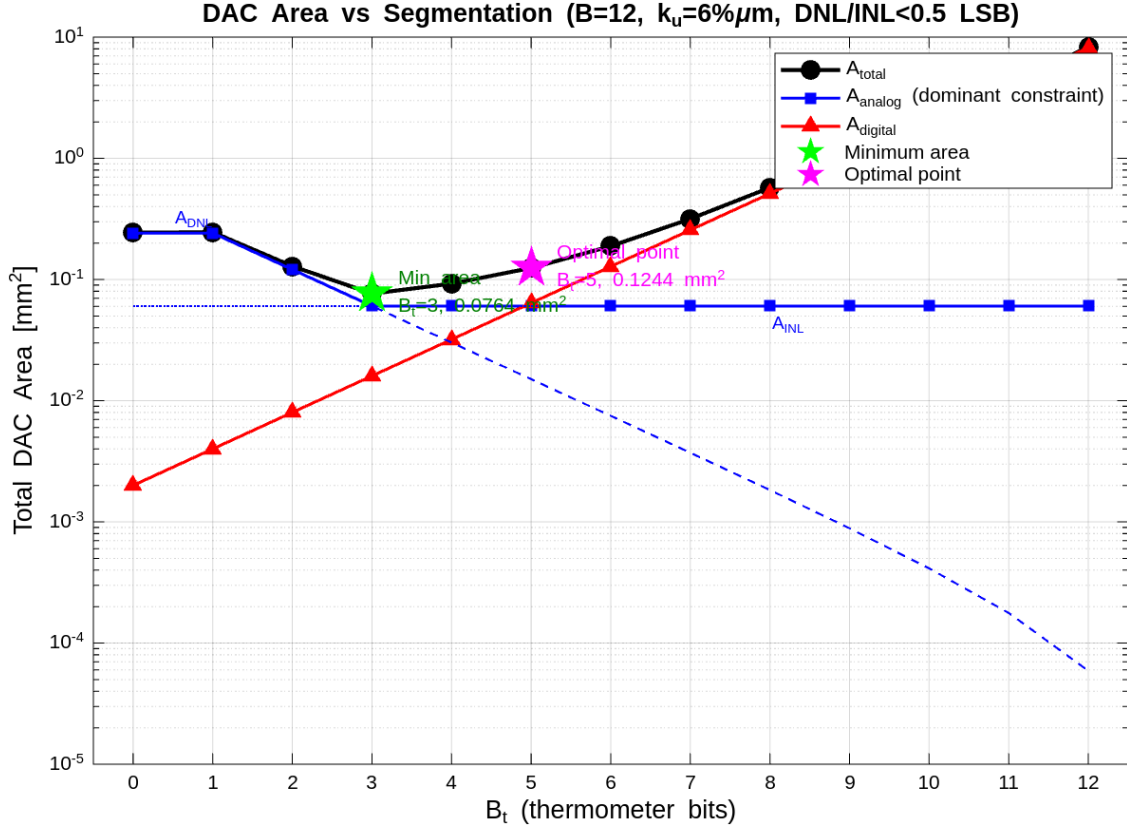


Figure 3: Total DAC area vs. number of thermometer bits B_t . The minimum area occurs at $B_t = 3$ (green), and the optimal point (analog $\approx$ digital) at $B_t = 5$ (magenta).

4 Monte Carlo Simulation

Task: Simulate the minimum-area design from Problem 3(a)(i) using the provided starter file `hw2_dac.m`, add DNL code, and iterate A_{unit} to achieve 95% yield.

4.1 Part (a): Added DNL Code

The following code was added to `hw2_dac.m` below the "Your code goes here" marker:

```
1 % calculate dnl
2 for m=1:r
3     avg_width = (code(m, end)-code(m, 1))/(2^B-1);
4     dnl(m, :) = diff(code(m,:)) ./ avg_width - 1;
5 end
6
7 % dnl scatter plot
8 figure(3); clf; hold on;
9 xlabel('Code');
10 ylabel('DNL [LSB]');
11 axis([0 2^B-2 -(dnlspec+0.05) (dnlspec+0.05)]);
12 line([0 2^B-2], [dnlspec dnlspec]);
13 line([0 2^B-2], [-dnlspec -dnlspec]);
14
15 bad_dacs_dnl=0;
16 for m=1:r
17     figure(3);
18     plot(0:2^B-2, dnl(m,:));
19     if find(abs(dnl(m,:))>dnlspec)
20         bad_dacs_dnl=bad_dacs_dnl+1;
21     end
22 end
23
24 figure(3); hold off;
25 title(sprintf('DNL envelope of %d runs. %d bad DAC(s).', ...
26     r, bad_dacs_dnl));
27
28 % dnl rms plot
29 dnl_rms = sqrt(sum( dnl.^2, 1 ) ./r);
30 [maxdnlrms dmax] = max(dnl_rms);
31 figure(4);
32 plot(0:2^B-2, dnl_rms, dmax, maxdnlrms, '*');
33 xlabel('Code');
34 ylabel('DNL [LSB]');
35 title(sprintf('RMS DNL of %d runs. (max=%1.3fLSBrms)', ...
36     r, maxdnlrms));
37 axis([0 2^B-2 0 maxdnlrms+0.01]);
```

4.2 Part (b): Initial Design Plots ($B_t = 3$, $A_{\text{unit}} = 14.75 \mu\text{m}^2$)

Using the minimum-area design from Problem 3(a)(i): $B_t = 3$, $A_{\text{unit}} = 14.75 \mu\text{m}^2$, $\sigma_u = 0.01562$.

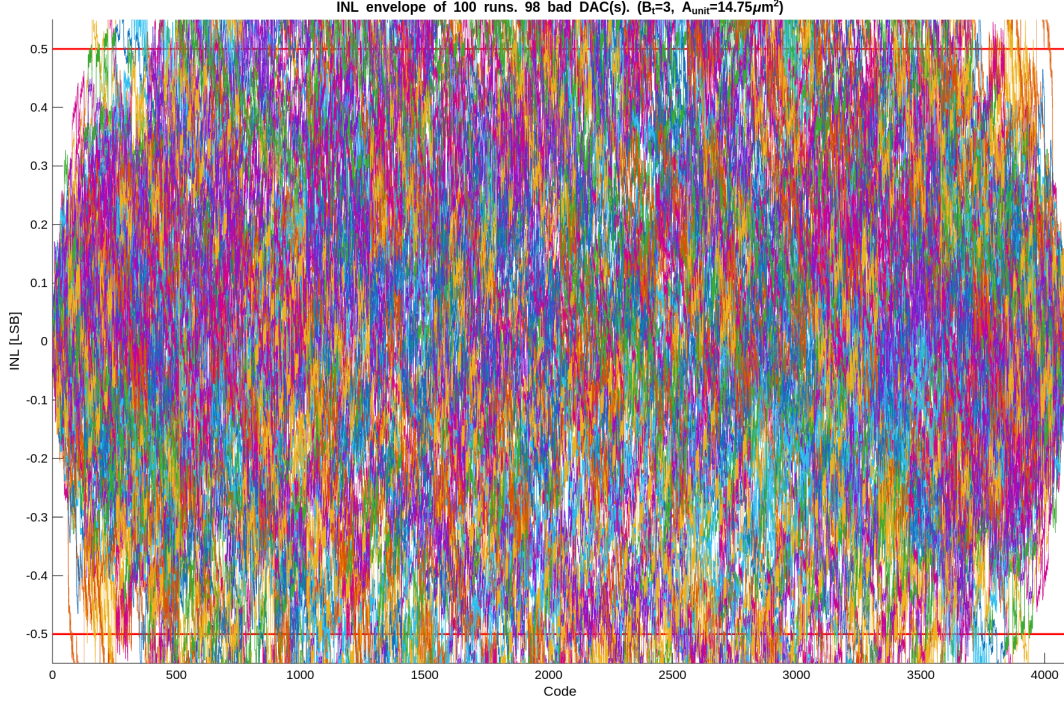


Figure 4: INL envelope of 100 Monte Carlo runs ($A_{\text{unit}} = 14.75 \mu\text{m}^2$). 98 bad DACs.

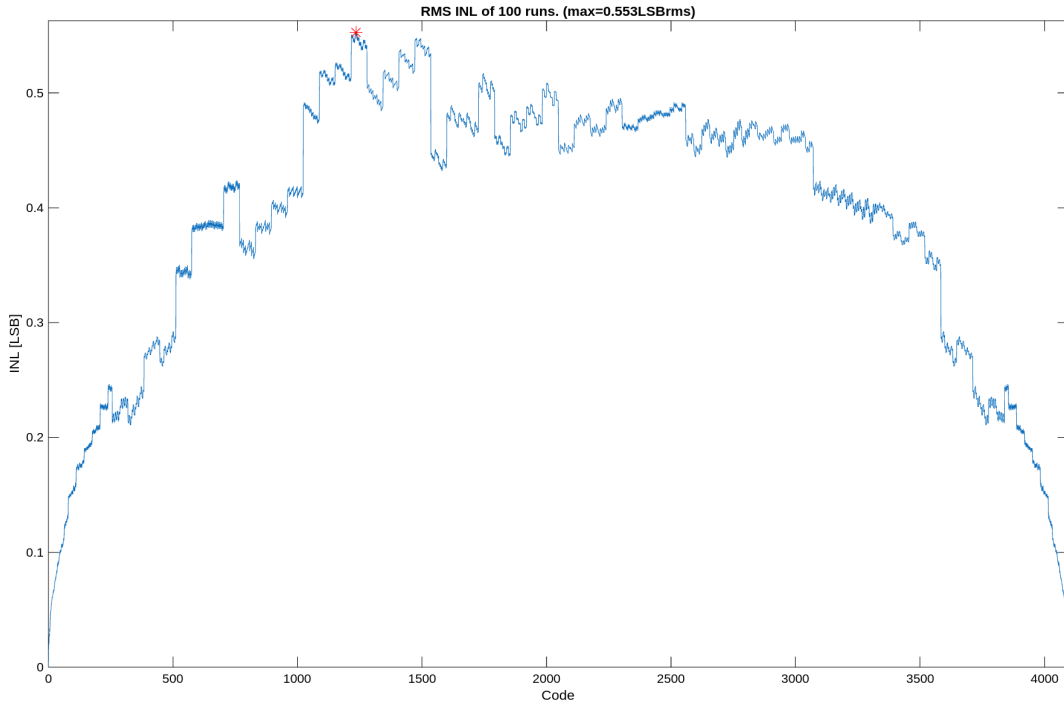


Figure 5: RMS INL of 100 runs ($A_{\text{unit}} = 14.75 \mu\text{m}^2$).

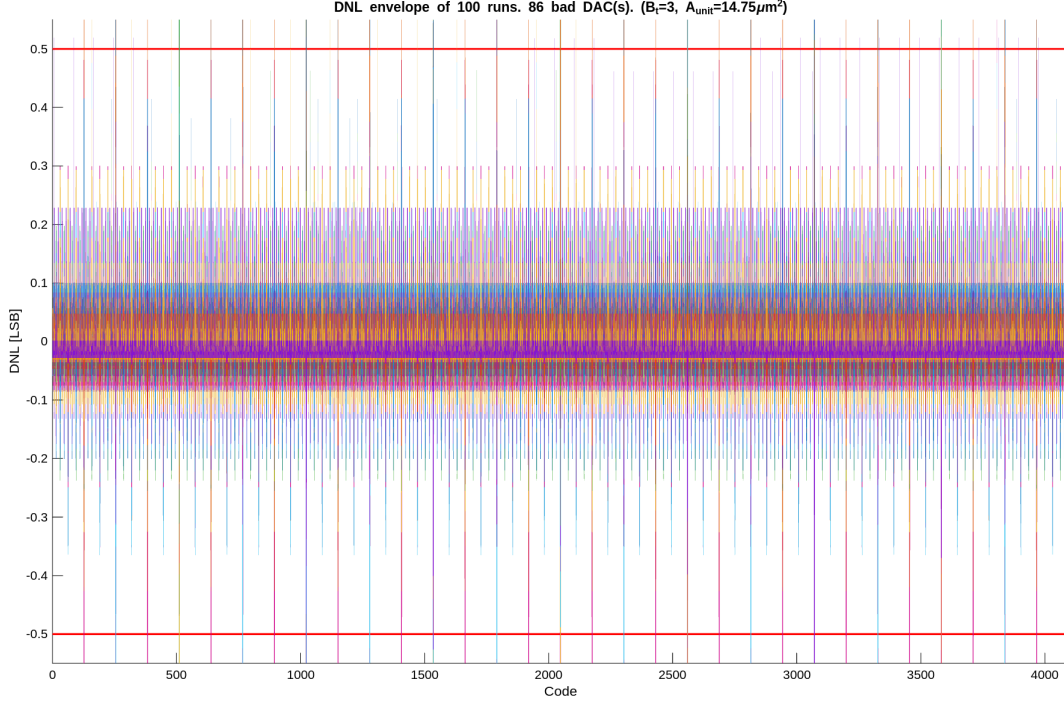


Figure 6: DNL envelope of 100 Monte Carlo runs ($A_{\text{unit}} = 14.75 \mu\text{m}^2$). 86 bad DACs.

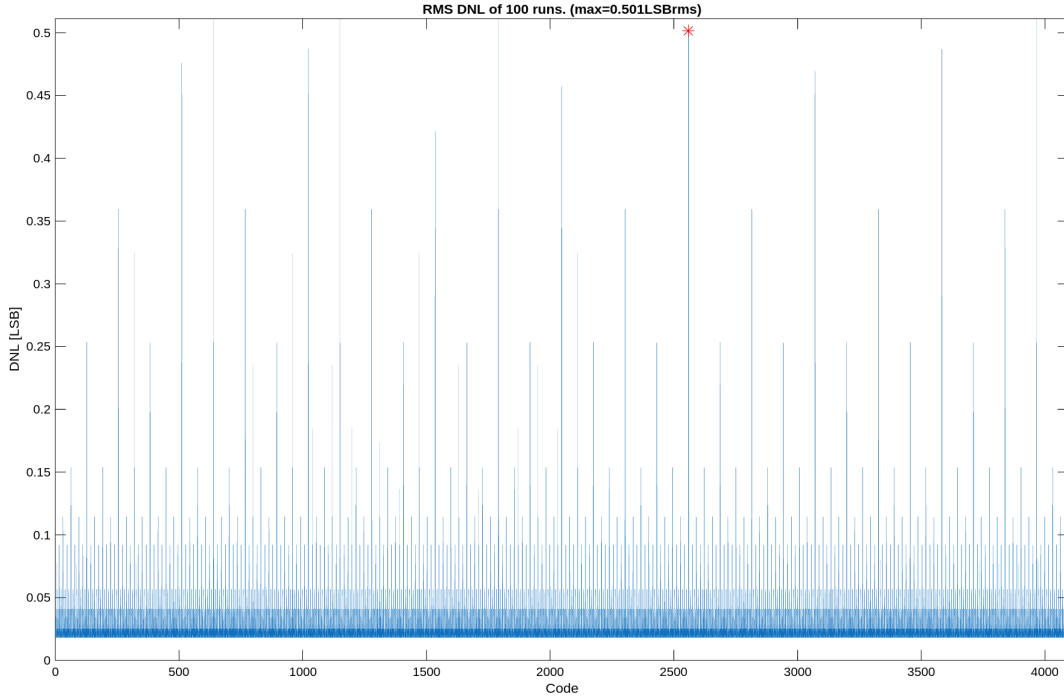


Figure 7: RMS DNL of 100 runs ($A_{\text{unit}} = 14.75 \mu\text{m}^2$).

Results: 98/100 bad INL, 86/100 bad DNL — yield is only 1%.

4.3 Part (c): Does the Initial Design Meet the Yield Spec?

No, the initial design completely fails the 95% yield target. Only 1 out of 100 DACs meets both $\text{DNL} < 0.5 \text{ LSB}$ and $\text{INL} < 0.5 \text{ LSB}$ simultaneously.

Why it fails: The hand calculation in Problem 3 followed the paper’s procedure of setting the spec equal to 1σ of the worst-case nonlinearity:

$$\sigma_{\text{INL,max}} = 32 \sigma_u = 0.5 \text{ LSB} \quad (1\sigma) \quad (10)$$

This means the worst-case INL exceeds 0.5 LSB about 32% of the time at the most critical code (midscale). With 4096 codes and many contributing to the peak INL, virtually every DAC has at least one code violating the spec.

To achieve 95% yield across all codes simultaneously, we need $\sim 3\sigma$ margin, i.e., the spec should equal approximately 3σ rather than 1σ . This requires σ_u to be $\sim 3\times$ smaller, meaning A_{unit} must be $\sim 9\times$ larger (since $A \propto 1/\sigma^2$).

4.4 Part (d): Adjusted Design for 95% Yield

Sweeping A_{unit} from 110 to $180 \mu\text{m}^2$ (with $B_t = 3$ fixed), averaging over 10 random seeds:

Table 4: Yield sweep results (averaged over 10 seeds, 100 runs each).

$A_{\text{unit}} (\mu\text{m}^2)$	σ_u	Average Yield
110	0.00572	89.7%
120	0.00548	91.8%
130	0.00526	93.7%
140	0.00507	95.5%
150	0.00490	96.1%
160	0.00474	97.2%

$A_{\text{unit}} = 140 \mu\text{m}^2$ achieves an average yield of **95.5%** (≤ 5 bad DACs on average out of 100), meeting the spec.

4.4.1 Final Design

Table 5: Final DAC design parameters.

Parameter	Value
B_t	3
B_b	9
A_{unit}	$140 \mu\text{m}^2$
σ_u	0.00507
A_{analog}	$4096 \times 140 = 573,440 \mu\text{m}^2$
A_{digital}	$8 \times 2000 = 16,000 \mu\text{m}^2$
A_{total}	$589,440 \mu\text{m}^2 = 0.589 \text{ mm}^2$

The final area is about $7.7\times$ larger than the 1σ minimum-area design (0.076 mm^2). The $\sim 9.5\times$ increase in unit element area corresponds to a $\sqrt{9.5} \approx 3.1\times$ reduction in σ_u , confirming that $\sim 3\sigma$ margin is needed for 95% yield with a 12-bit DAC.

4.4.2 Final Plots ($B_t = 3$, $A_{\text{unit}} = 140 \mu\text{m}^2$)

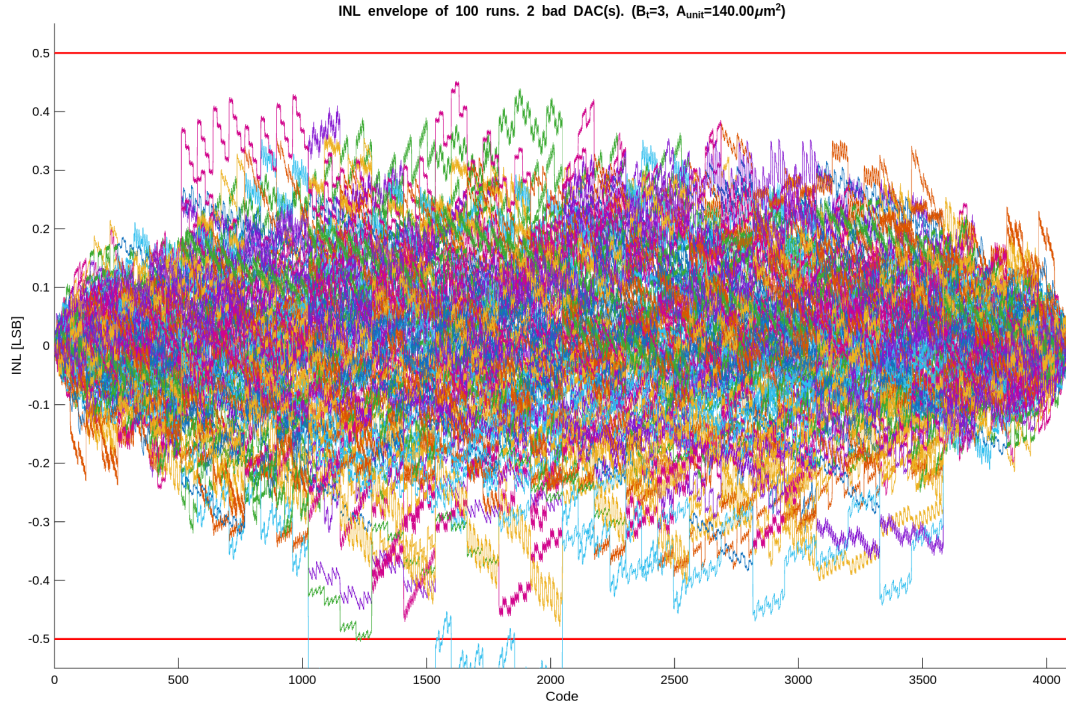


Figure 8: INL envelope of 100 Monte Carlo runs ($A_{\text{unit}} = 140 \mu\text{m}^2$). 2 bad DACs.

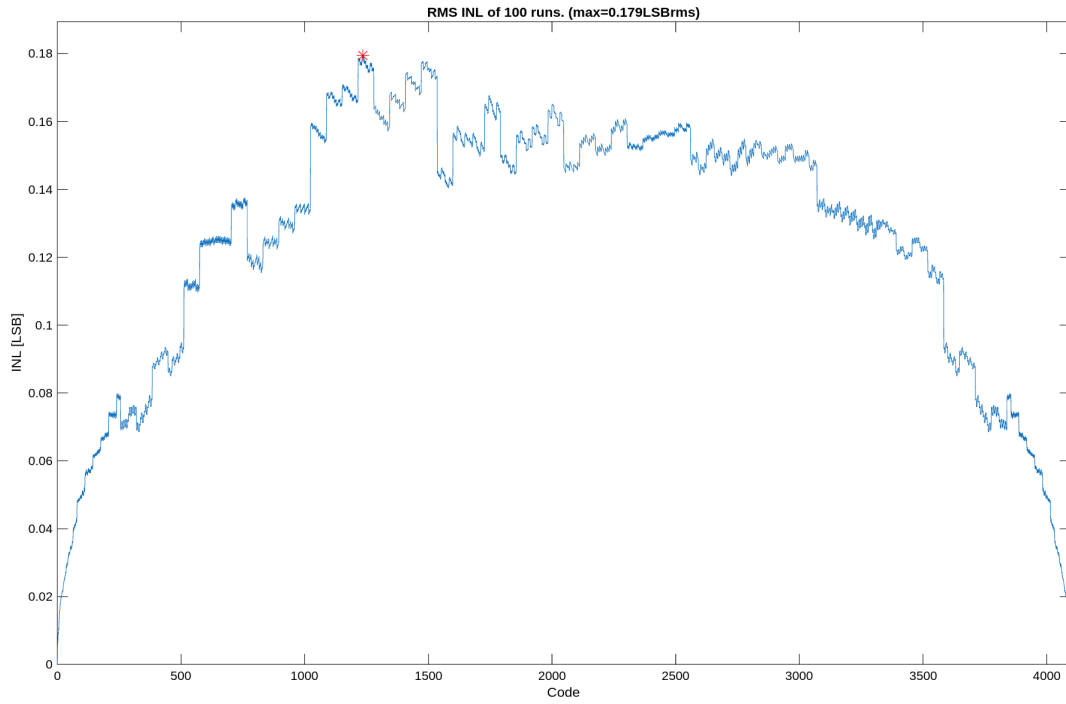


Figure 9: RMS INL of 100 runs ($A_{\text{unit}} = 140 \mu\text{m}^2$). Peak = 0.179 LSBrms at midcode.

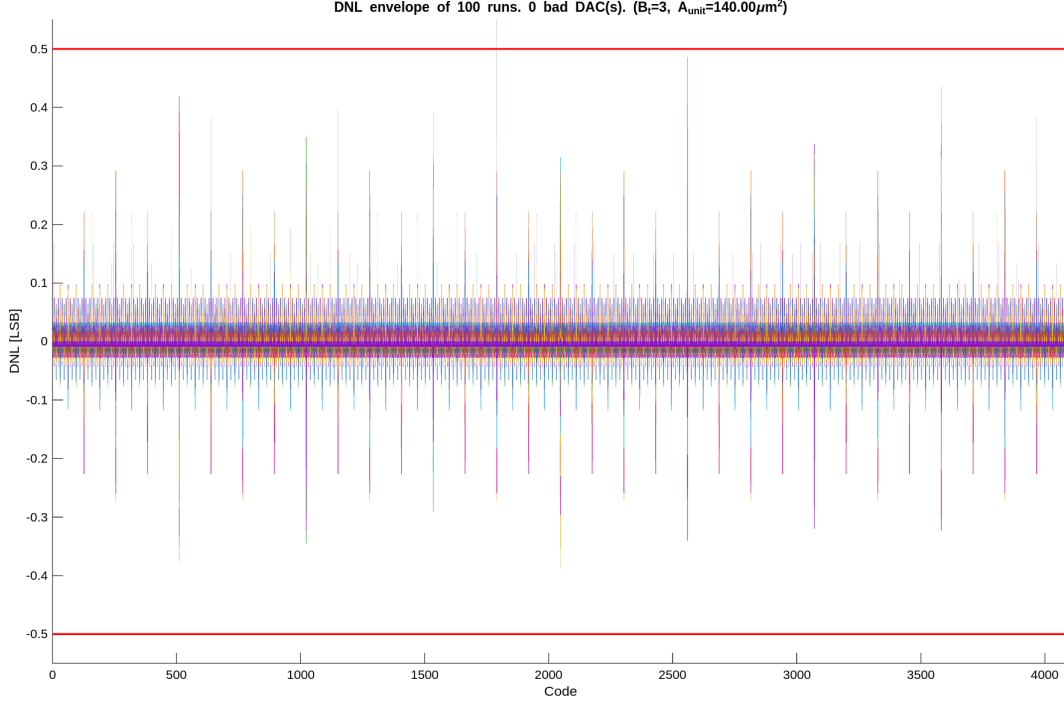


Figure 10: DNL envelope of 100 Monte Carlo runs ($A_{\text{unit}} = 140 \mu\text{m}^2$). 0 bad DACs.

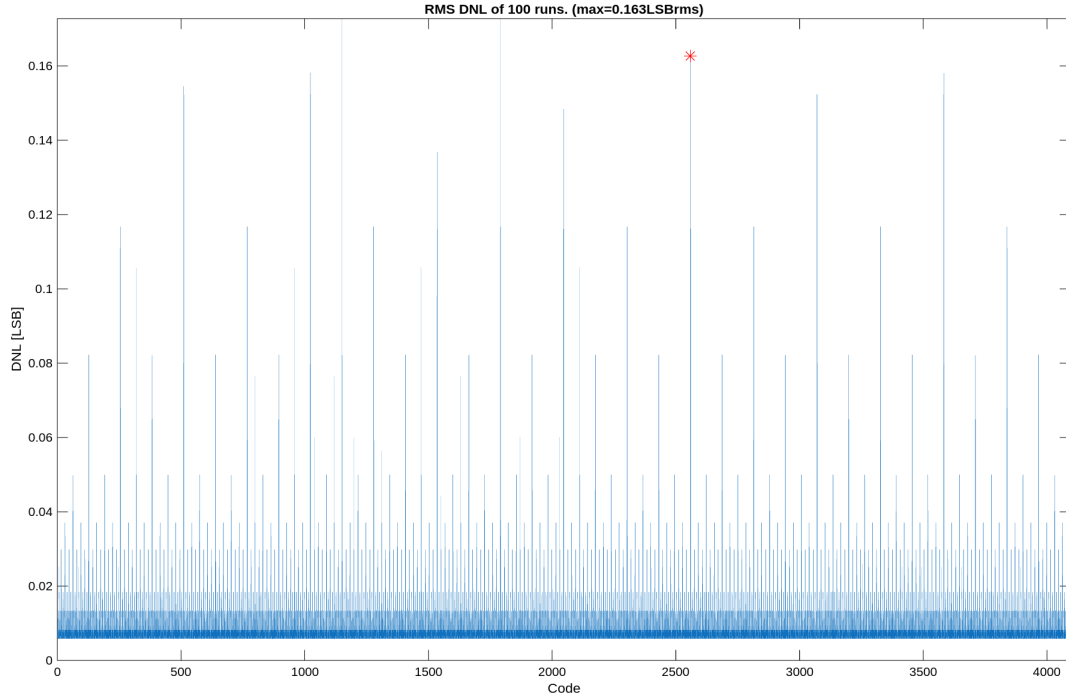


Figure 11: RMS DNL of 100 runs ($A_{\text{unit}} = 140 \mu\text{m}^2$). Peak = 0.163LSBrms at major carry transitions.

Final results: Bad INL: 2/100, Bad DNL: 0/100, Overall yield: **98%** — meets the 95% target.